

W. A. SANUJI DEWMINI WEERASINGHE

Data Science Undergraduate | University of Peradeniya

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PROFESSIONAL PROFILE

Final-year Data Science undergraduate at the University of Peradeniya with a strong foundation in statistics, machine learning, and data engineering. Experienced in building end-to-end data solutions and deep learning models, with hands-on research in time-series forecasting. Seeking a Data Science / Data Engineering internship to leverage analytical skills and contribute to intelligent automation or business intelligence solutions.

EDUCATION

BSc (Hons) in Data Science — University of Peradeniya

2022 – Present

GPA: 3.45 / 4.00

G.C.E. Advanced Level

2020

- Combined Mathematics: A · Chemistry: B · Physics: C.

WORK EXPERIENCE

Bank of Ceylon — Job Skills Development Trainee

Jan – Apr 2021

Ratmalana Super Grade Branch · School Leavers Programme · Front & Back Office Operations

- Completed the Bank's structured Job Skills Development Training Programme for School Leavers, gaining hands-on experience in day-to-day commercial-bank operations.
- Supported Front-Office and Back-Office banking operations, with a focus on customer service.

PROJECTS

Monsoon-Stratified Rooftop Solar PV Forecasting for Grid Stability

2026 – Present

Final-Year Research Project (DSC 4996) — University of Peradeniya

Python · PyTorch · CNN-LSTM & Transformer · ResUNet / YOLOv8-seg · Satellite Imagery & GIS · Time-Series Forecasting

- Built a satellite-imagery pipeline across 5 Sri Lankan districts that detects rooftop solar panels, benchmarking YOLOv8-seg and ResUNet segmentation models to estimate district-level rooftop PV potential.
- Benchmarked CNN-LSTM against a Transformer for monsoon-stratified day- and week-ahead PV forecasting (5 districts × 4 monsoon regimes), evaluated with MAE, RMSE, R^2 , and skill scores against a persistence baseline.
- Engineered a leakage-safe, 39-feature time-series dataset spanning 8 years and 5 districts from NASA POWER and public meteorological sources, applying strict chronological splits and bootstrap confidence intervals.

Retrieval-Augmented Q&A System (RAG)

2026

Python · ChromaDB · Sentence-Transformers · Streamlit

- Built an end-to-end RAG application answering natural-language questions grounded in a custom document corpus, with source citations to minimise hallucination.
- Implemented document chunking, vector embedding, and semantic retrieval; deployed an interactive Streamlit interface.

Customer Churn Prediction with MLOps

2026 – Present

Python · XGBoost · MLflow · FastAPI · Docker · GitHub Actions

- Developed an XGBoost churn-prediction model with feature engineering, class-imbalance handling, and SHAP-based explainability.
- Productionised the model with MLflow experiment tracking, a FastAPI/Docker serving layer, CI/CD via GitHub Actions, and data-drift monitoring.

Tea Leaf Disease Classification

2025

Python · PyTorch · VGG16 / ResNet50 · scikit-learn · Matplotlib

- Applied transfer learning with VGG16 and ResNet50 (ImageNet-pretrained) to classify three tea leaf diseases (Brown Blight, Algal Leaf, White Spot), using data augmentation (flip, rotation, colour jitter) to improve generalisation.
- Benchmarked both architectures with accuracy, precision, recall, and F1-score, analysing training curves and confusion matrices to select the stronger model.

PHM Tweet Classification

2025

Python · TensorFlow / Keras · LSTM · Bi-LSTM · scikit-learn

- Compared LSTM and Bi-LSTM architectures for Public Health Monitoring tweet classification on imbalanced social-media text.
- Addressed class imbalance with class weighting and threshold optimisation, validating performance through cross-validation.

AI-Based Medical Claims Validation System

2025

Python · Django REST Framework · React (TypeScript) · OpenCV

- Built a full-stack, multi-portal web application (Branch, Digital, and Admin portals) for end-to-end insurance claim submission, AI-powered document analysis, and fraud detection.
- Developed a weighted five-factor AI risk-scoring engine — date validity, policy-limit check, name matching, prescription–bill gap, and document integrity — to auto-approve, flag, or reject claims; integrated fraud detection logic with role-based JWT authentication across Branch, Digital, and Admin portals.

Business Intelligence Framework for Sales Performance

2024

Power BI · SQL · ETL · Data Warehousing · OLAP

- Designed an end-to-end BI framework consolidating multi-branch retail data into a centralised warehouse via ETL pipelines.
- Built an OLAP cube and interactive Power BI dashboards visualising KPIs, sales trends, and branch comparisons.

TECHNICAL SKILLS

Languages: Python · SQL · R · Java

ML & Deep Learning: Pandas · NumPy · scikit-learn · PyTorch · TensorFlow / Keras · XGBoost

Modelling & NLP: CNN · LSTM / Bi-LSTM · Transformers · Transfer Learning · RAG · Embeddings · ChromaDB

Computer Vision & Geospatial: Image Segmentation (ResUNet, U-Net, YOLOv8-seg) · OpenCV · Satellite Imagery · GIS · OCR

Data Engineering & MLOps: ETL · Data Warehousing · PostgreSQL · MySQL · Docker · MLflow · FastAPI · GitHub Actions

BI, Visualisation & Tools: Power BI · Streamlit · Matplotlib · Seaborn · Git · GitHub · Jupyter

SOFT SKILLS

Leadership · Teamwork · Effective Communication · Critical Thinking · Problem Solving · Analytical Reasoning · Time Management

ACHIEVEMENTS & AWARDS

- Champion — IEEE INSL Central Provincial New Innovation Competition (Hope Project), 2024.
- Champion — KickStart Weekend 2024, SIREN Business Idea Competition (Business Idea Pitch), 2024.
- Research & Development Director — DataEX Society, University of Peradeniya, 2026.

REFERENCES

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